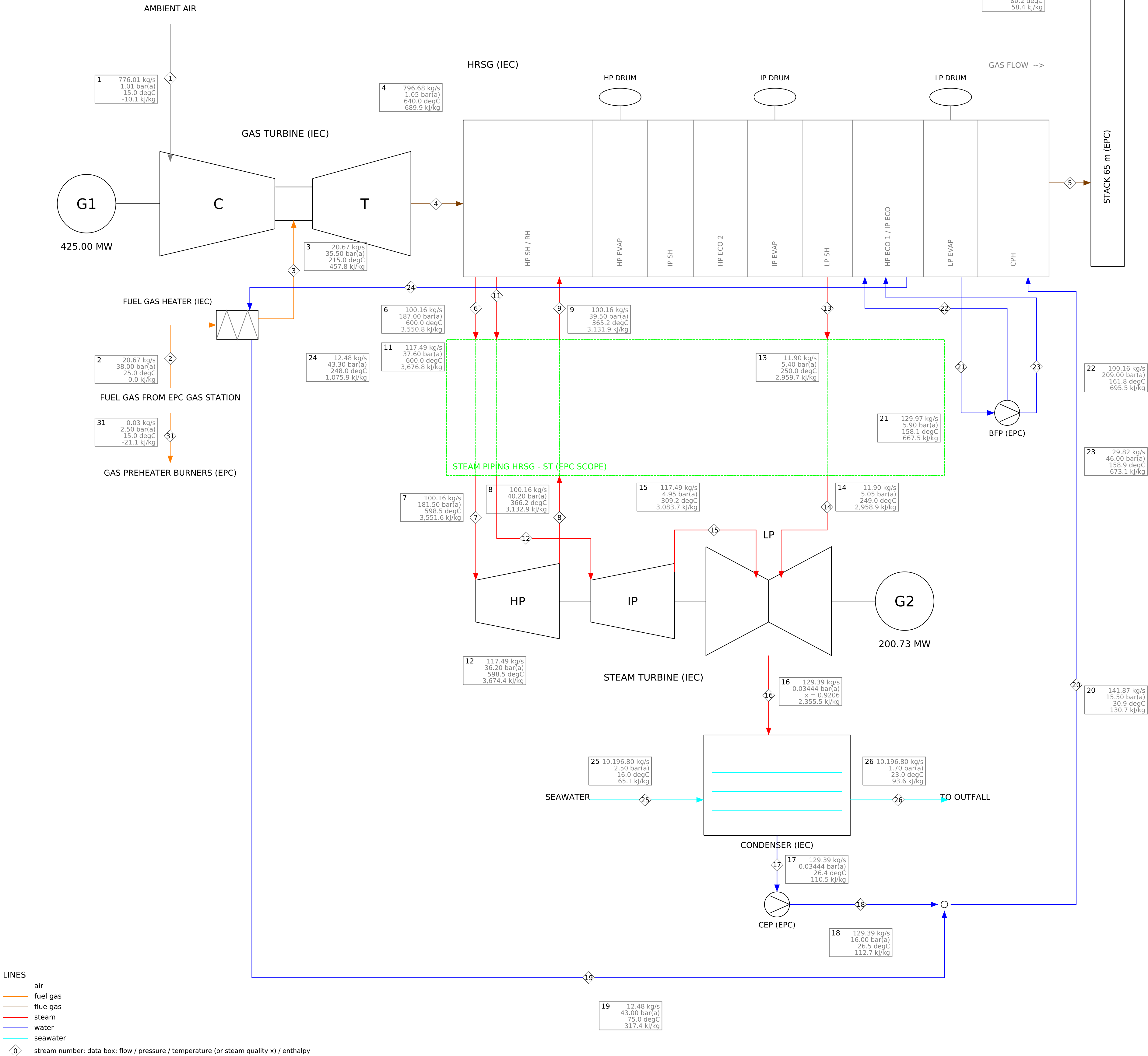




STREAM TABLE - SRC-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	776.01	1.01	15.0	-10.1	-7.8
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	20.67	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	20.67	35.50	215.0	457.8	9.5
4	GT exhaust gas to HRSG inlet	796.68	1.05	640.0	689.9	549.6
5	Flue gas at HRSG outlet to stack	796.68	1.01	80.2	58.4	46.5
6	HP steam at HRSG outlet (HP SH outlet)	100.16	187.00	600.0	3,550.8	355.6
7	HP steam at ST main stop valve inlet	100.16	181.50	598.5	3,551.6	355.7
8	Cold reheat steam at HP turbine exhaust	100.16	40.20	366.2	3,132.9	313.8
9	Cold reheat steam at HRSG reheater inlet	100.16	39.50	365.2	3,131.9	313.7
10	IP superheated steam (joins cold reheat inside the HRSG)	17.34	40.30	330.0	3,041.7	52.7
11	Hot reheat steam at HRSG outlet	117.49	37.60	600.0	3,676.8	432.0
12	Hot reheat steam at ST IP stop/control valve inlet	117.49	36.20	598.5	3,674.4	431.7
13	LP steam at HRSG outlet	11.90	5.40	250.0	2,959.7	35.2
14	LP steam at ST LP admission	11.90	5.05	249.0	2,958.9	35.2
15	IP turbine exhaust (crossover, internal)	117.49	4.95	309.2	3,083.7	362.3
16	LP turbine exhaust to condenser (UEEP)	129.39	0.0344	x 0.921	2,355.5	304.8
17	Condensate at hotwell outlet / CEP suction	129.39	0.0344	26.4	110.5	14.3
18	Condensate at CEP discharge	129.39	16.00	26.5	112.7	14.6
19	Fuel gas heater water return to condensate	12.48	43.00	75.0	317.4	4.0
20	Condensate to HRSG condensate preheater inlet	141.87	15.50	30.9	130.7	18.5
21	Feedwater from LP drum to boiler feed pumps	129.97	5.90	158.1	667.5	96.8
22	HP feedwater at BFP discharge to HP economiser	100.16	209.00	161.8	695.5	69.7
23	IP feedwater at BFP interstage to IP economiser	29.82	46.00	158.9	673.1	20.1
24	IP economiser water to performance fuel gas heater	12.48	43.30	248.0	1,075.9	13.4
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	65.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	23.0	93.6	954.4
31	Fuel gas to the station preheater burners (water-bath heaters)	0.03	2.50	15.0	-21.1	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	425.00	MW
ST output (generator terminals)	200.73	MW
Gross output	625.73	MW
Auxiliary loads	-9.86	MW
Transformer losses (GSU, UAT, MV/LV)	-1.93	MW
NET OUTPUT (HV terminals)	613.94	MW
Heat input (LHV)	983.02	MW
Gross heat rate (LHV)	5,647	kJ/kWh
NET HEAT RATE (LHV)	5,764	kJ/kWh
Net efficiency (LHV)	62.45	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	101,752	Sm3/h
Circulating water flow	36,112	m3/h
GT load	100.00	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.004 MW (tolerance 0.25 MW)
OK	energy balance COND: residual +0.027 MW (tolerance 0.97 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual -0.018 MW (tolerance 0.98 MW)
OK	energy balance HRSG: residual +2.039 MW vs stated casing loss 2,012 MW (tolerance 0.97 MW)
OK	energy balance PIPE: residual +0.298 MW (tolerance 1.14 MW)
OK	energy balance ST: residual +0.000 MW (tolerance 0.82 MW)
OK	fuel LHV from the composition (ISO 6976) 47.48 MJ/kg vs DP-CASE-0018 47.48 MJ/kg (-0.00 %)
OK	auxiliary + transformer losses 1.88 % of gross vs DP-CASE-0031 <= 2.2 %
OK	condenser pressure 34.44 mbar(a) vs DP-OWNR-0021 <= 36.0 mbar(a)
OK	gas flow 101,752 Sm3/h = 84.8 % of DP-OWNR-0033 (120,000 Sm3/h)
OK	net export 613.9 MW vs DP-OWNR-0051 connection capacity 640 MW (this case)
OK	gross output 625.73 MW vs IG-01 >= 615.00 (margin +1.74 %)
OK	gross heat rate 5,646.88 kJ/kWh vs IG-02 <= 5,735.00 (margin +1.54 %)
OK	GT output 425.00 MW vs IG-03 >= 418.00 (margin +1.67 %)
OK	condenser pressure 34.44 mbar(a) vs IG-04 <= 35.50 (margin +2.99 %)
OK	net output 613.94 MW vs PG-01 >= 600.00 (margin +2.32 %)
OK	net heat rate 5,764.23 kJ/kWh vs PG-02 <= 5,860.00 (margin +1.63 %)
OK	CW temperature rise 7.00 K vs PG-13 <= 7.00 (margin +0.00 %)
OK	Guarantee cover: at IG-01/IG-02 limits and this HMB's auxiliaries the net plant is 603.2 MW / 5,847 kJ/kWh vs Owner PG-01 600 MW / PG-02 5,860 kJ/kWh

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-NG-100
CASE	Site Reference Conditions, natural gas, base load (guarantee case)
DOCUMENT	ALP-HMB-SRC-NG-100
STATUS	preliminary - generated by engine hmb v1.1.1
DATA	b6f4b950c7f3b7e0 2026-09-29
NOTE	all values calculated from the database; do not edit